



**UG Program in Electronics and Communication  
(Advanced Communication Technology)**

**Experiment No. 5**

**Generation (and detection) of Binary ASK**

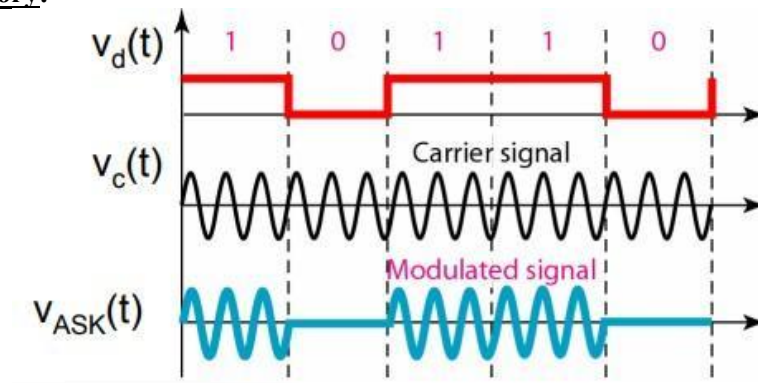
**5.1 Aim:** To study the Binary Amplitude Shift Keying Generation (and detection) technique and plot the modulated and demodulated waveforms.

**5.2 Course Outcome:** Compare the performances of different digital modulation techniques

**5.3 Learning Objectives:** Students will be able

- To understand the principles of Binary Amplitude Shift Keying (ASK) modulation and demodulation techniques.
- To generate and detect Binary ASK signals and plot the modulated and demodulated waveforms.

**5.4 Related Theory:**



**BASK Modulation waveforms**

Binary Amplitude Shift Keying (BASK) is the digital modulation technique. In amplitude shift keying, the amplitude of the carrier signal is varied to create signal elements. Both frequency and



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phase remain constant while the amplitude changes. In BASK, the amplitude of the carrier assumes one of the two amplitudes dependent on the logic states of the input bit stream. The modulated signal is given by the following expression,

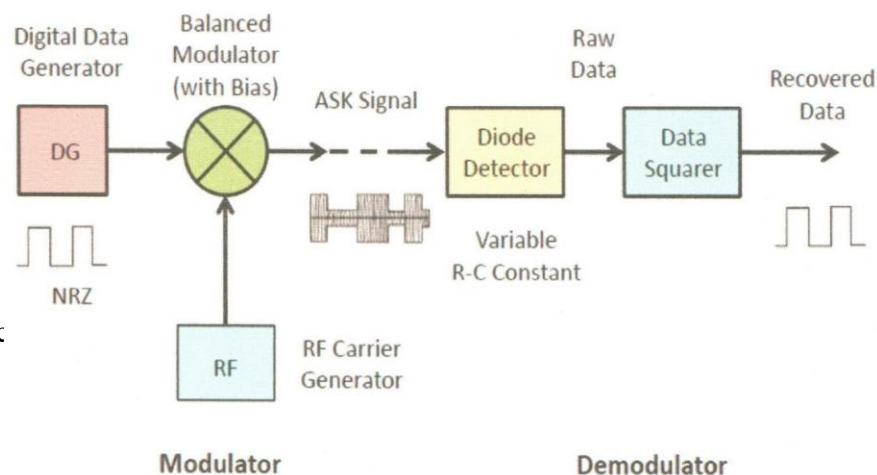
$$V_{ASK} = 0, \quad \text{for symbol '0'}$$

$$V_{ASK} = A \cos \omega_c t \quad \text{for symbol '1'}$$

In the above waveform  $V_d(t)$  is the data generator output i.e., unipolar NRZ data,  $V_c(t)$  is the carrier signal and  $V_{ASK}(t)$  is the modulated waveform. There are sharp discontinuities shown at the transition points. These result in the signal having an unnecessarily wide bandwidth. Band limiting is generally introduced before transmission, in which case these discontinuities would be 'rounded off'. The band limiting may be applied to the digital message, or the modulated signal itself. The data rate is often made a sub-multiple of the carrier frequency.

A block diagram of the BASK modulation demodulation system consists of the following sections.

1. Digital Data Generator Section
2. RF Carrier Oscillator
3. ASK Modulator section
4. ASK Demodulator Section
5. Power supply.





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**Data Generator Section:** This section generates the digital data signal. It consists of many digital ICs. The carrier clock of 1.28MHz is generated by IC74HC04. This carrier clock is divided by 16 to produce Bit Clock (Bk) of 80 KHz. This Bk is divided by 8 to generate Word Clock(wk) of 10KHz. Then the word of 8-bit data is generated using IC74166 and eight data toggle switches. By pressing different toggle switches different data word can be generated. This data word is called as NRZ Data. This NRZ data is then given to ASK Modulator.

**R.F. Carrier Oscillator section:** BC548 is used to generate RF sine wave signal. 15K POT is used to vary its frequency. The frequency range is 100 KHz to 1MHz. Then transistor pair BC558/BC548 and BC558/548 are used to amplify this RF oscillation. The amplified RF sine wave signal available at emitter junction of BC558/BC548 is given to RF carrier terminal through amplitude preset and POT. There are two RF Carrier signals. One is directly given to RF I/P terminal of balanced modulator section. Second RF signal can be connected at RF I/P of Product Detector section. 10K POT is used to vary the amplitude of sine wave signal. The output amplitude varies from 0 to 10Vpp.

**ASK Modulator section:** IC 1496 is used as balanced modulator. The modulating NRZ signal is connected at pin 1 through buffer transistor BC548B. This IC has two inputs as it works as balanced modulator. The second input can be connected at pin 4 through buffer transistor BC548B. The RF carrier signal is connected at pin 8 through coupling capacitor from RF carrier oscillator section. The modulated outputs are available at pin 12 and 6 of this IC which are then balanced amplified by transistor pairs BC548B and BC558B. The final balanced modulated output is available at ASK o/p terminal. 100K preset is used to balance carrier signal while 1K preset is used to balance input audio signal. Output 1K preset is used to adjust output zero Dc level. Output 2K2 preset is used to vary level of ASK modulated signal.



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**ASK Demodulator using Diode detector:** This circuit consists of diode OA79 and capacitors C1,C2 and load resistor R. It works as an envelope detector circuit. R & C form Low pass filter which reduces the carrier frequency ripple in the output. The values of R & C are chosen to remove even the modulating signal frequency components. C1 and C2 can be selected through switch.

**Power Supply section:** Following output D.C. voltages are required to operate BASK Modulation demodulation system.

+15V, 250mA,

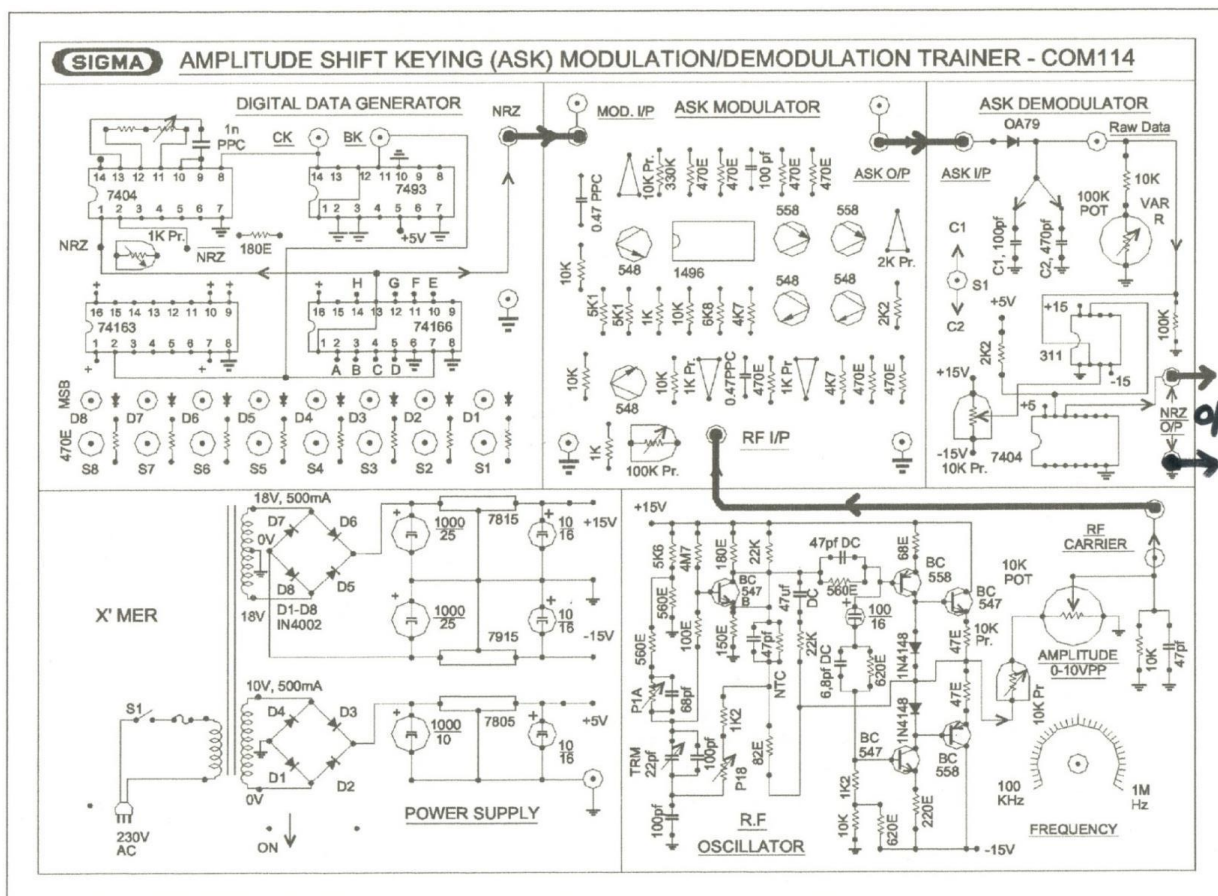
-15V, 250mA,

+5V, 250mA

IC 7805 for +5V, 7815 for =15V and 7915 for -15V, these ICs are supplied different dc input voltages by two bridge rectifiers consisting of D1-D4 and D5-D8.

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**Circuit Connection Diagram**



**BASK Circuit Connection Diagram**



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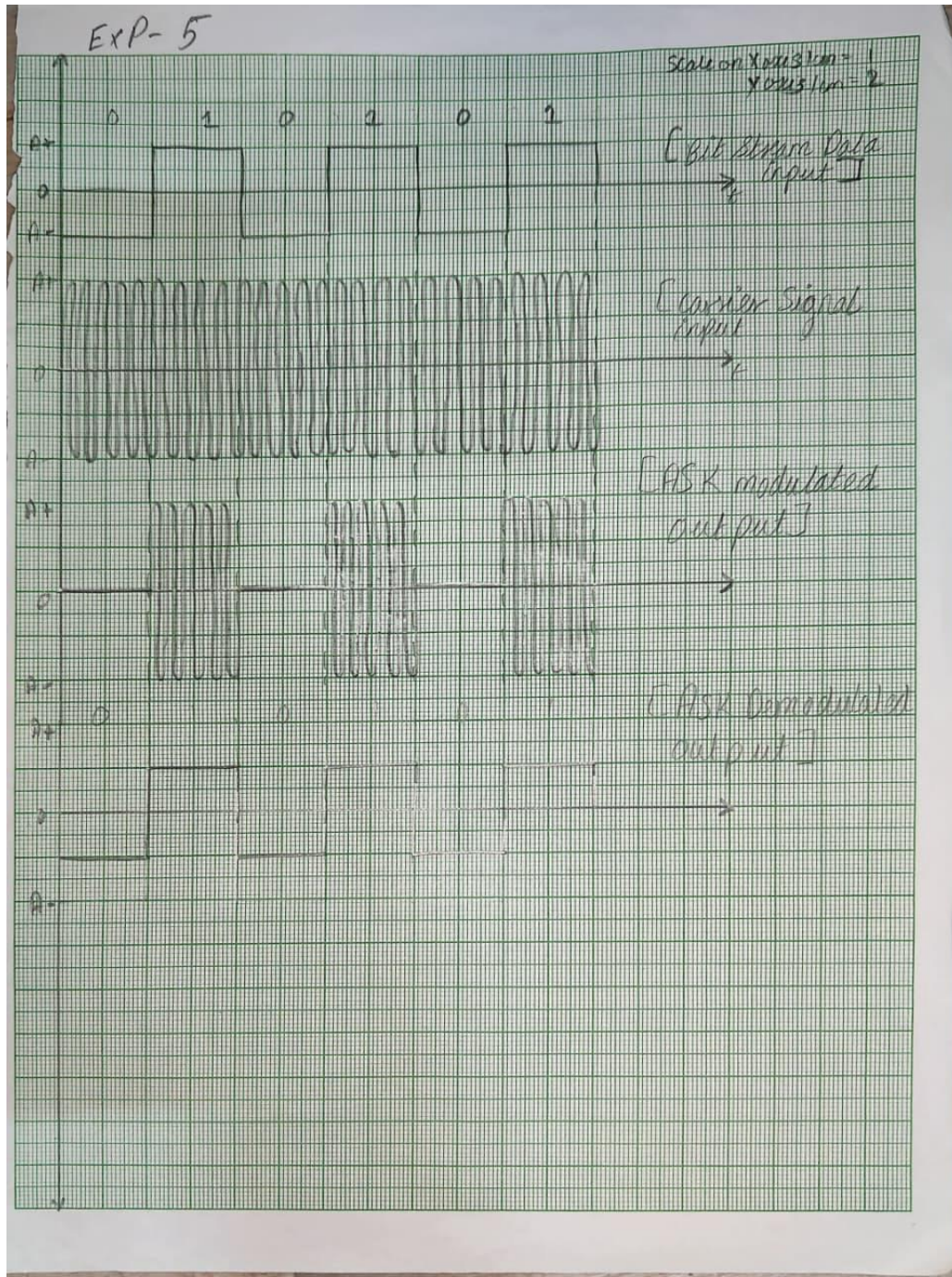
### **Procedure:**

1. Switch on the trainer kit.
2. Observe Clock signal on C.R.O. Measure the time and amplitude of Clock signal.
3. Toggle the switches to give data input to ASK Modulator section and observe the NRZ data on the channel 1 of the C.R.O. Measure time and amplitude of NRZ signal.
4. Connect Channel 2 of C.R.O. to RF O/P terminal of RF oscillator and measure the time and amplitude of carrier signal.
5. Apply 8-bit digital data by toggling the data generator switches to ASK Modulator and observe the ASK modulated waveform on the C.R.O.
6. By varying the switches observed the change in modulated waveform.
7. Vary RF Carrier oscillator frequency and amplitude and observe the effect on modulated waveform.
8. Now Connect BASK output to the demodulator section.
9. Observe the demodulated Raw NRZ data at the output of the detector circuit.
10. Vary the R.F. Oscillator Frequency and Amplitude. Also vary the RC time constant in the diode detector by selecting different capacitors C1,C2 by switch and by varying POT and observing raw data.
11. Observe the NRZ data at the output of data squarer and check whether it is same as NRZ data acts as an input to the modulation circuit.
12. Plot all the waveforms on the Graph paper



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**5.5 Waveforms:**





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### **5.6 Conclusion:**

In this experiment, Binary Amplitude Shift Keying (ASK) modulation and demodulation were successfully studied and implemented. The ASK signal was generated by varying the carrier amplitude according to the binary input data, and accurate detection of the original message was achieved. The modulated and demodulated waveforms clearly demonstrated the working principle of Binary ASK. This study helps in comparing ASK with other digital modulation techniques in terms of simplicity and performance.

### **5.7 Questions:**

#### **1. List Advantages and disadvantages of BASK**

##### **Advantages:**

- Simple to generate and demodulate.
- Requires less complex circuitry compared to other digital modulation schemes.
- Easy implementation using basic analog components.
- Suitable for low-cost and low-data-rate communication systems.

##### **Disadvantages:**

- Highly susceptible to noise and interference.
- Poor power efficiency since information is carried in amplitude.
- Requires higher signal power for reliable transmission.
- Not suitable for long-distance or high-speed communication.

#### **2. List real time applications of BASK**

- Optical communication systems (On–Off Keying).
- RFID systems (low-frequency and passive RFID).
- Infrared remote control systems.
- Low-data-rate wireless sensor networks.
- Simple digital communication and laboratory experiments.